

Lecture 7 In-Class Worksheet: Should a Firm Raise Its Price?

Setup

A firm wants to **maximize profit**. It currently charges a single price P for its good.

Assume:

- The firm charges **one price** (no price discrimination).
- **Marginal cost is positive:** $MC > 0$.
- The firm considers raising its price **a little** (a small increase in P).

Recall:

$$\pi = TR - TC$$

$$TR = P \cdot Q(P)$$

Part A — Warm-up: What changes when price rises?

When the firm raises price slightly:

1. Quantity demanded Q (**rises / falls**).
2. Total revenue TR may (**rise / fall**) depending on demand elasticity.
3. Total cost TC changes because output changes and $MC > 0$.

Key idea: If the firm sells fewer units, it saves production costs on those units.

Part B — Elastic Demand

Suppose demand at the current price is **elastic**.

Question 1: Revenue

If demand is elastic, a small increase in price causes quantity to fall by a **larger percentage** than the price rises.

- Does $TR = P \cdot Q$ increase or decrease?

Explain briefly:

Question 2: Cost

When price rises, quantity falls, so the firm produces fewer units.

- Does total cost TC increase or decrease?

Why does the assumption $MC > 0$ matter here?

Question 3: Profit

Profit is given by:

$$\pi = TR - TC$$

Given your answers above, when demand is elastic and the firm raises price a little, what happens to profit π ? Explain briefly:

Part C — Inelastic Demand

Suppose demand at the current price is **inelastic**.

Question 4: Revenue

If demand is inelastic, a small increase in price causes quantity to fall by a **smaller percentage** than the price rises.

- Does $TR = P \cdot Q$ increase or decrease?

Explain briefly:

Question 5: Cost

When price rises, quantity falls, so the firm produces fewer units.

- Does total cost TC increase or decrease?

TC (increases / decreases)

Question 6: Profit

Profit is given by:

$$\pi = TR - TC$$

Given your answers above, when demand is inelastic and the firm raises price a little:

π (increases / decreases / ambiguous)

Explain briefly:

Part D — Synthesis

Fill in the rule

- If demand is **elastic**, raising price a little tends to _____ profit.
- If demand is **inelastic**, raising price a little tends to _____ profit.

One-sentence intuition

Complete the sentence:

Raising price is more attractive when demand is _____ because revenue moves _____, and costs move _____ when output falls (since $MC > 0$).